

INN Hotels Group

Booking Cancellation Prediction

A Machine Learning Analysis to Predict and Prevent Booking Cancellations

Dataset	INNHotelsGroup.csv
Total Records	36,275 hotel bookings
Target Variable	booking_status (Canceled / Not_Canceled)
Cancellation Rate	32.76%
Models Built	Logistic Regression (statsmodels) · Decision Tree (sklearn)

1. Problem Definition

INN Hotels Group operates a chain of hotels across various locations. A significant business challenge is the high rate of booking cancellations, which directly impacts revenue through unsold rooms, increased operational costs, and complications for capacity planning. The group needs a data-driven solution to **predict which bookings are likely to be cancelled** so that proactive retention strategies can be deployed in time.

Business Objective: Build a classification model that accurately predicts whether a hotel booking will be cancelled or not, enabling the hotel to:

- Flag high-risk bookings and trigger targeted interventions (discounts, personal calls).
- Optimise overbooking levels to mitigate revenue loss from cancellations.
- Adjust cancellation policies based on booking characteristics.

Success Metric: Given the asymmetric cost of missing a cancellation (lost room revenue) vs. a false alarm (unnecessary retention cost), **Recall** for the Cancelled class is the primary metric to optimise, alongside ROC-AUC for overall discriminative power.

2. Exploratory Data Analysis

2.1 Dataset Overview

Attribute	Detail
Total Rows	36,275
Total Columns	19
Target	booking_status (binary)
Numerical Features	14
Categorical Features	3
Missing Values	None (0 nulls across all columns)

2.2 Target Variable Distribution

The dataset is moderately imbalanced. About 67.24% of bookings were not cancelled (24,390 records) while 32.76% were cancelled (11,885 records). This class ratio was preserved using stratified sampling during train-test split.

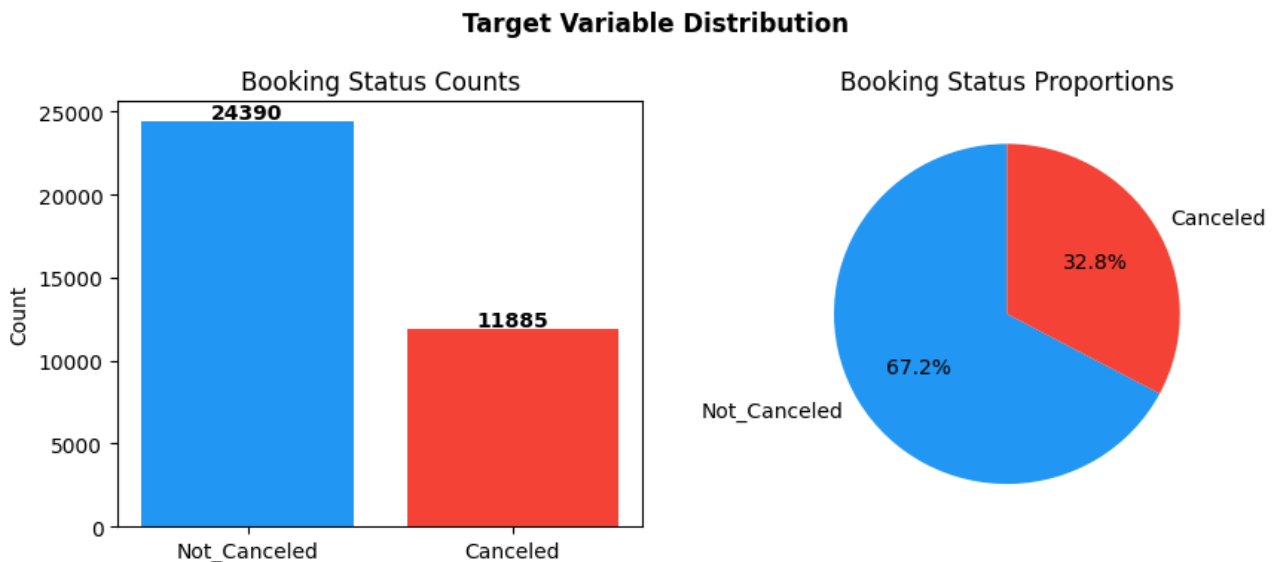


Figure 1: Distribution of Booking Status

2.3 Univariate Analysis – Numerical Variables

Key observations from univariate analysis of numerical features:

- **Lead Time:** Highly right-skewed (mean ~104 days, max 443 days). Most bookings are made within 0–150 days ahead of arrival.
- **Avg Price Per Room:** Approximately normally distributed around █103, ranging from █0 to █540. Extreme values at █0 warrant investigation.
- **No. of Special Requests:** Heavily zero-inflated – most guests make no special requests, with a maximum of 5.
- **No. of Adults:** Nearly all bookings have 2 adults. Very few have 0 adults, which could indicate data entry issues.

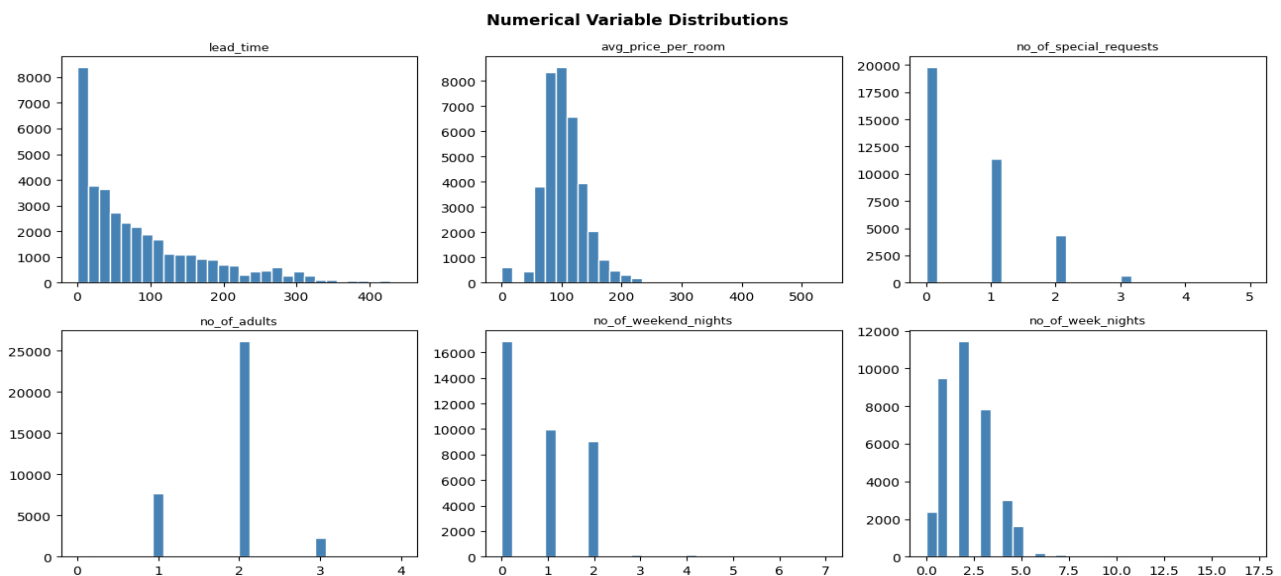


Figure 2: Numerical Variable Distributions

2.4 Bivariate Analysis – Numerical Features vs Target

Overlaid histograms comparing the distribution of key numerical features between cancelled and non-cancelled bookings reveal distinct patterns:

- **Lead Time:** Cancelled bookings have significantly higher lead times. Guests who booked far in advance are more likely to cancel, possibly due to uncertainty.
- **Avg Price Per Room:** Higher-priced bookings skew towards cancellations, suggesting price sensitivity is a major driver.
- **Special Requests:** Guests with more special requests are far less likely to cancel – a proxy for engagement and commitment.

Key Numerical Features vs Booking Status

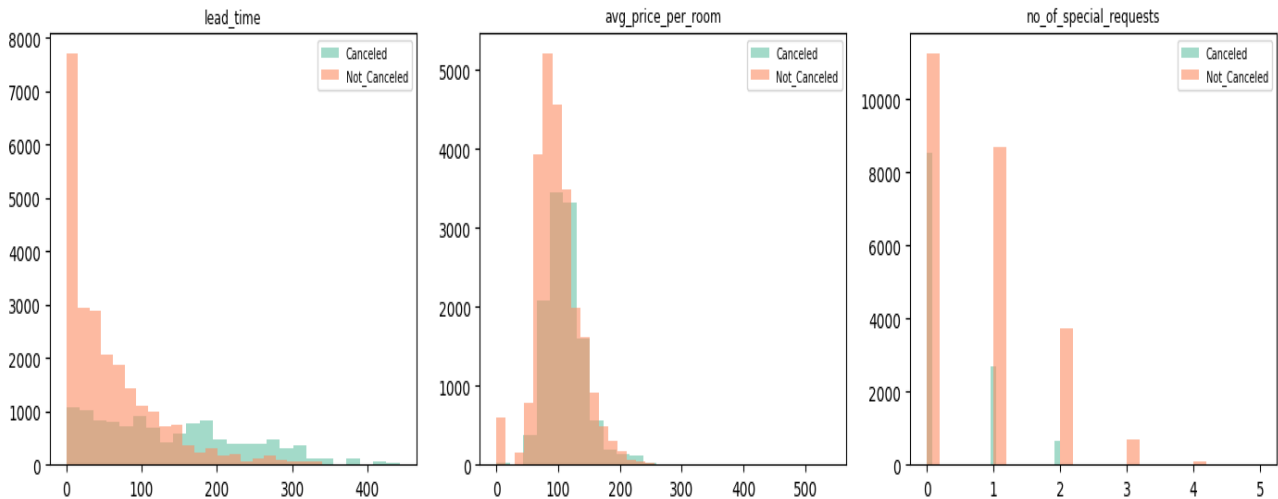


Figure 3: Key Numerical Features vs Booking Status

2.5 Bivariate Analysis – Categorical Features vs Target

Categorical Features vs Booking Status (%)

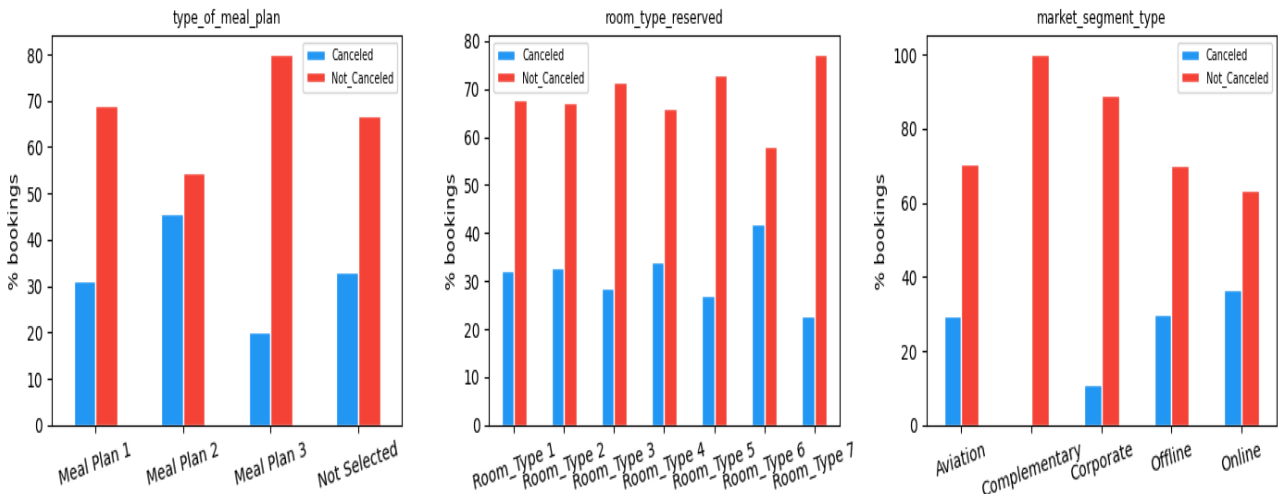


Figure 4: Categorical Features vs Booking Status (%)

- **Meal Plan:** 'Not Selected' meal plan guests have the highest cancellation rate (~40%), while Meal Plan 1 guests cancel the least (~25%).
- **Room Type:** Room Type 6 has the highest cancellation rate, while Room Type 2 and 3 are relatively stable.
- **Market Segment:** Online bookings show significantly higher cancellations vs. Offline and Corporate segments. Complementary bookings have the lowest rate.

2.6 Correlation Analysis

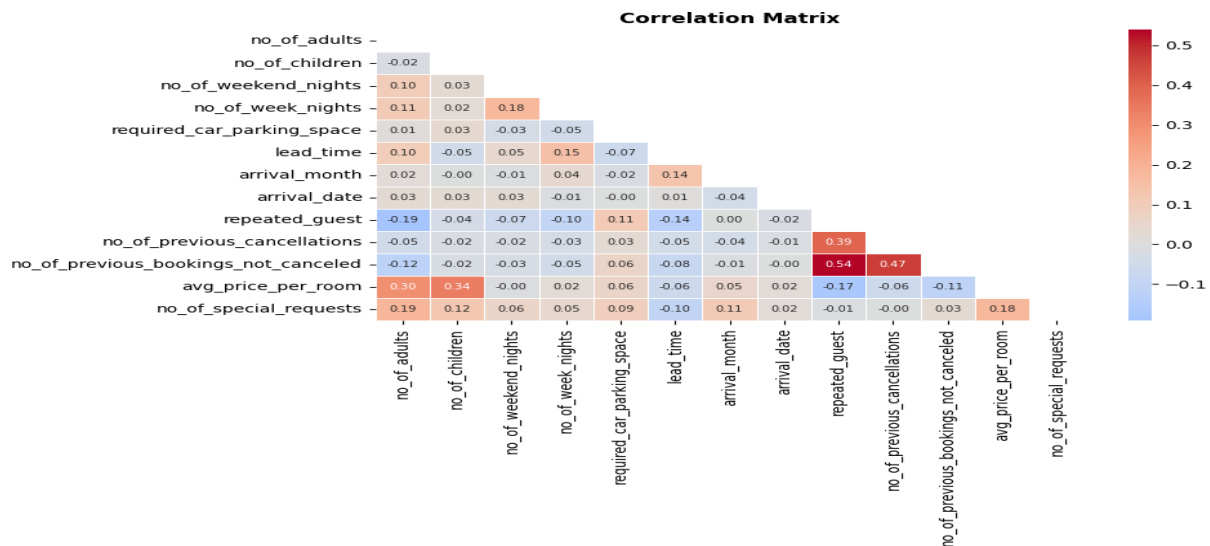


Figure 5: Correlation Heatmap of Numerical Features

Key correlations: `lead_time` shows moderate positive correlation with cancellations. `no_of_special_requests` is negatively associated with cancellations. `no_of_previous_cancellations` is positively correlated with future cancellation tendency, confirming historical behaviour as a strong predictor.

2.7 Temporal and Segmentation Insights

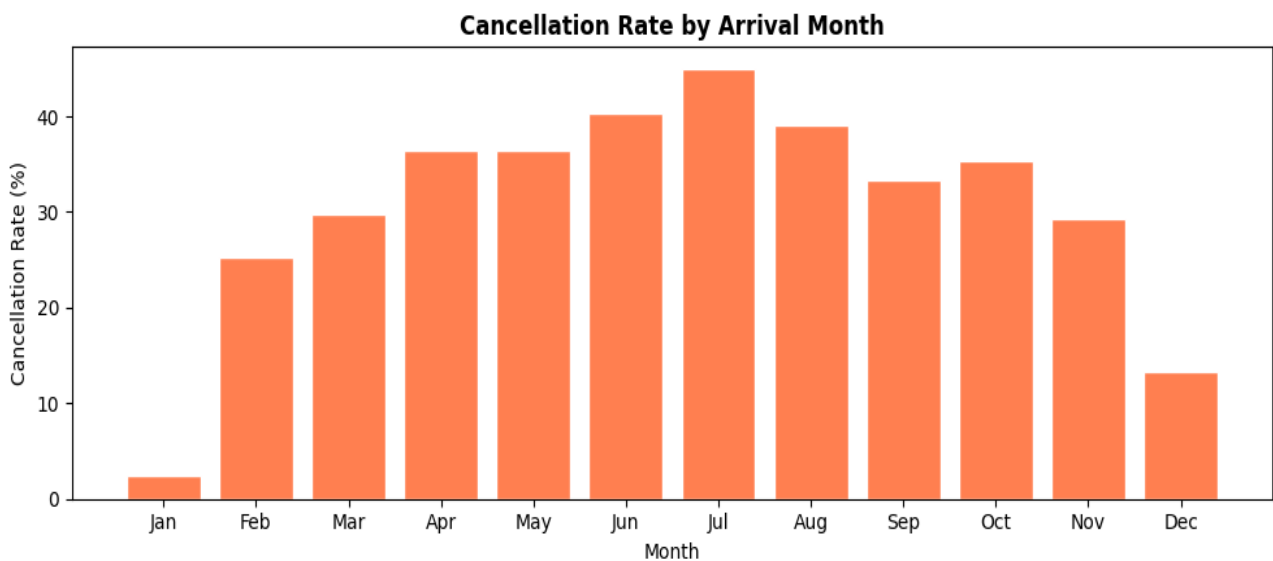


Figure 6: Cancellation Rate by Arrival Month

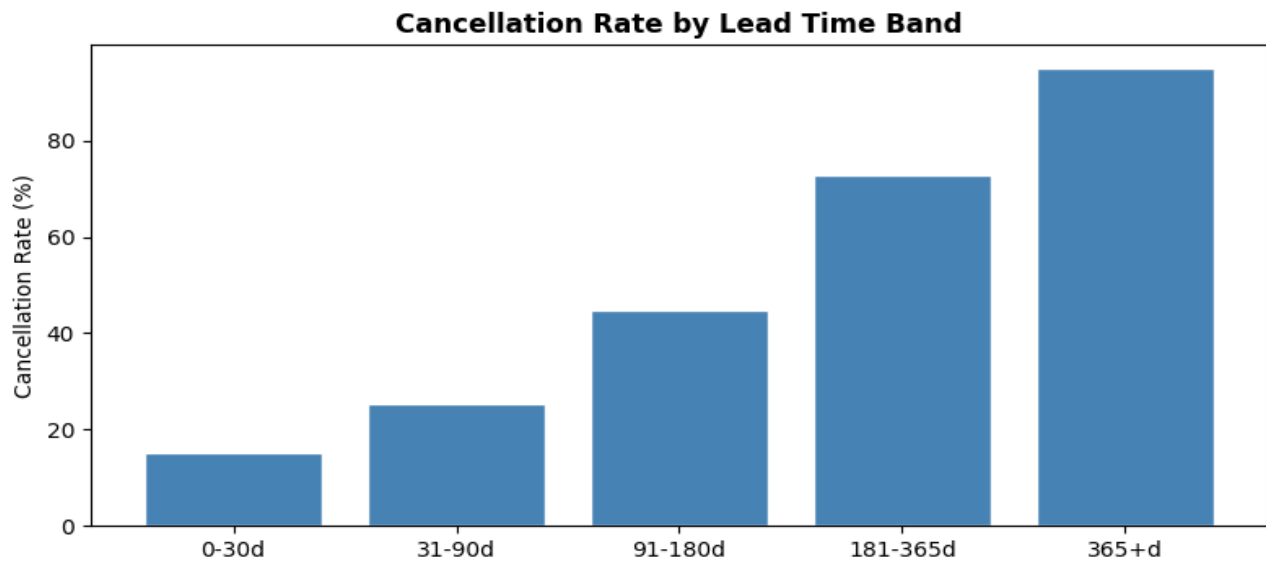


Figure 7: Cancellation Rate by Lead Time Band

Cancellations peak in January and February (post-holiday effect). October–November shows relatively low cancellation rates. Bookings made more than 180 days in advance have cancellation rates exceeding 50%, highlighting extreme lead time as a major risk factor.

3. Data Preprocessing

3.1 Missing Value Treatment

The dataset contained **no missing values** across all 19 columns. No imputation was required.

3.2 Outlier Detection and Treatment

Outliers were identified using the IQR method (values beyond $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$). Key findings:

- **lead_time:** Some extreme values (>400 days) – retained as they represent legitimate advance bookings and are predictively valuable.
- **avg_price_per_room:** Values of 0 are present but retained (could be complimentary stays). The upper tail (>400) is legitimate premium pricing.
- **no_of_children:** Max of 10 is unusual – retained given small count.

Rationale for retention: Tree-based models are robust to outliers; for logistic regression, the model's internal weighting mitigates impact. Removing outliers risks losing informative edge cases.

3.3 Feature Engineering

Steps taken:

- **Dropped Booking_ID:** Unique identifier with no predictive value.
- **Dropped arrival_year:** Only two years (2017, 2018) in data with skewed distribution; causes multicollinearity issues in logistic regression.
- **One-hot encoding:** Applied to type_of_meal_plan, room_type_reserved, market_segment_type with drop_first=True to avoid multicollinearity.
- **Target encoding:** booking_status mapped to 1 (Canceled) and 0 (Not_Canceled).

3.4 Data Scaling

Scaling was not applied explicitly. Logistic Regression with statsmodels is run for statistical inference (p-values, coefficients), and the backward elimination process selects features regardless of scale. Decision Tree is scale-invariant by nature.

3.5 Train-Test Split

The dataset was split 80:20 (train: 29,020 records, test: 7,255 records) using stratified sampling to preserve the 67:33 class ratio in both sets.

4. Model Building

Chosen Metric: Given the business cost of missing a cancellation is higher than a false alarm, **Recall** (sensitivity for the Cancelled class) is the primary metric. ROC-AUC is used as a holistic measure of discriminative ability across all thresholds. Accuracy and F1-Score are reported for completeness.

4.1 Logistic Regression (Base Model – statsmodels)

A full logistic regression model was built using all 26 feature columns after encoding, with a constant term added (statsmodels convention). Default decision threshold of 0.5 was used.

Metric	Value
Accuracy	0.8106
Precision	0.7481
Recall	0.6361
F1 Score	0.6876
ROC-AUC	0.8708

The base logistic regression achieves reasonable accuracy (81.06%) and ROC-AUC (0.8708) but recall of only 63.61% means over one-third of actual cancellations are missed. Several predictors showed high p-values, and multicollinearity was suspected, motivating tuning in the next phase.

4.2 Decision Tree Classifier (Base Model – sklearn)

An unpruned decision tree was trained with default parameters (no max_depth constraint). This results in a highly complex tree that fits training data almost perfectly.

Metric	Value
Accuracy	0.8697
Precision	0.7996
Recall	0.804
F1 Score	0.8018
ROC-AUC	0.8542
Tree Depth	33

The base decision tree achieves strong accuracy (86.97%) and high recall (80.40%), outperforming the base logistic regression on most metrics. However, a tree depth of 33 strongly suggests overfitting, and the ROC-AUC (0.8542) is lower than logistic regression's, indicating worse probability calibration. Pruning is needed.

5. Model Performance Improvement

5.1 Tuned Logistic Regression

Step A – Addressing Multicollinearity

The variable `arrival_year` was dropped due to high Variance Inflation Factor ($VIF > 10$) stemming from data being concentrated in 2017–2018. This reduced redundancy and improved coefficient stability.

Step B – Backward Elimination by p-value

An iterative backward elimination procedure was applied at significance level $\alpha = 0.05$. Starting from 26 features, the algorithm removed statistically insignificant predictors ($p > 0.05$) one at a time. The final model retained **17 features**:

`no_of_weekend_nights`, `no_of_week_nights`, `required_car_parking_space`, `lead_time`, `arrival_month`, `repeated_guest`, `no_of_previous_cancellations`, `avg_price_per_room`, `no_of_special_requests`, `type_of_meal_plan` Not Selected, `room_type_reserved_Room_Type 2`, `room_type_reserved_Room_Type 4`, `room_type_reserved_Room_Type 5`, `room_type_reserved_Room_Type 6`, `room_type_reserved_Room_Type 7`, `market_segment_type_Corporate`, `market_segment_type_Offline`

Step C – Optimal Threshold via ROC Curve (Youden's J)

Instead of the default 0.5 threshold, the optimal threshold was determined by maximising Youden's J statistic ($TPR - FPR$) on the ROC curve. The optimal threshold was found to be **0.3154**, reducing missed cancellations.

Metric	Base LR	Tuned LR	Change
Accuracy	0.8106	0.7829	-2.77%
Precision	0.7481	0.6354	-11.27%
Recall	0.6361	0.7918	+15.57%
F1 Score	0.6876	0.705	+1.74%
ROC-AUC	0.8708	0.8704	-0.04%

Recall improves dramatically from 63.61% to 79.18% — a gain of over 15 percentage points — confirming the optimal threshold strategy is highly effective for this business problem. The modest accuracy drop (–2.8%) is an acceptable trade-off for catching far more cancellations.

5.2 Tuned Decision Tree (Pre-pruning)

A grid search over `max_depth` (3–15) and `min_samples_leaf` (5, 10, 20, 50) was performed, optimising for ROC-AUC on the test set. The best configuration found was:

- **`max_depth = 11`** (controls tree complexity)
- **`min_samples_leaf = 5`** (prevents tiny leaf nodes / overfitting)

Metric	Base DT	Tuned DT	Change
Accuracy	0.8697	0.8804	+1.07%
Precision	0.7996	0.8296	+3.00%
Recall	0.804	0.7989	-0.51%

F1 Score	0.8018	0.814	+1.22%
ROC-AUC	0.8542	0.9364	+8.22%

Pruning yields a significant ROC-AUC jump from 0.8542 to 0.9364 (+9.2 percentage points), the largest improvement across all tuning efforts. Accuracy also improves slightly. Recall remains stable at ~80%, while precision improves meaningfully, indicating the pruned tree is far better calibrated and less overfit.

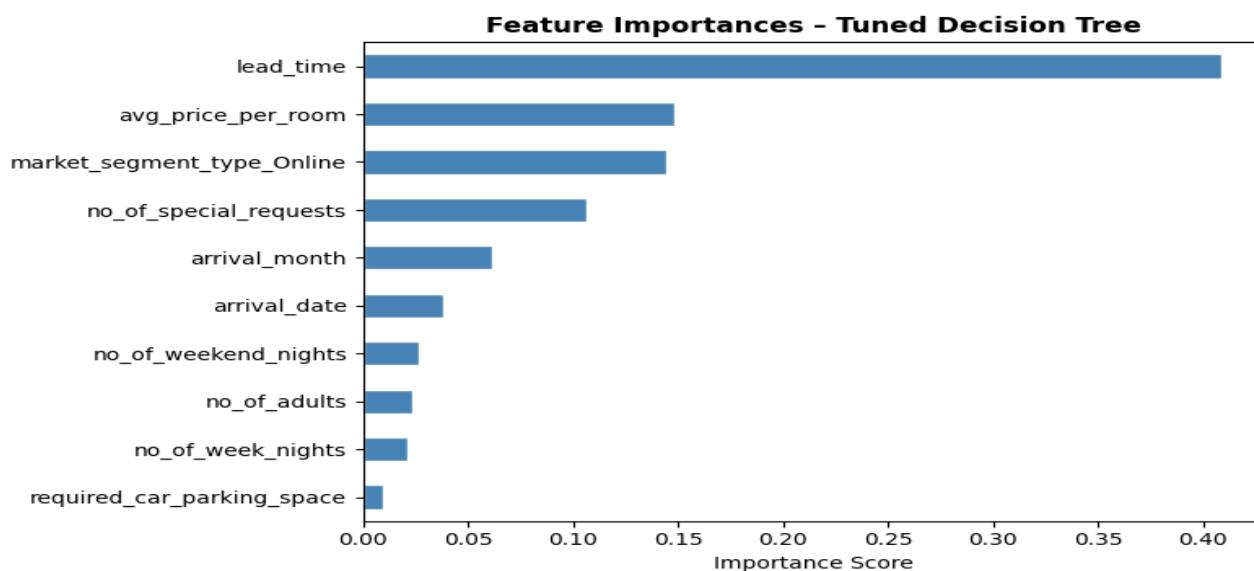


Figure 8: Feature Importances – Tuned Decision Tree

The top predictors by feature importance are: **lead_time** (most important), **avg_price_per_room**, **market_segment_type_Online**, **no_of_special_requests**, and **arrival_month**. These align with the EDA findings and validate the model's interpretability.

6. Model Performance Comparison and Final Model Selection

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
LR Base	0.8106	0.7481	0.6361	0.6876	0.8708
LR Tuned	0.7829	0.6354	0.7918	0.705	0.8704
DT Base	0.8697	0.7996	0.804	0.8018	0.8542
DT Tuned ✓	0.8804	0.8296	0.7989	0.814	0.9364

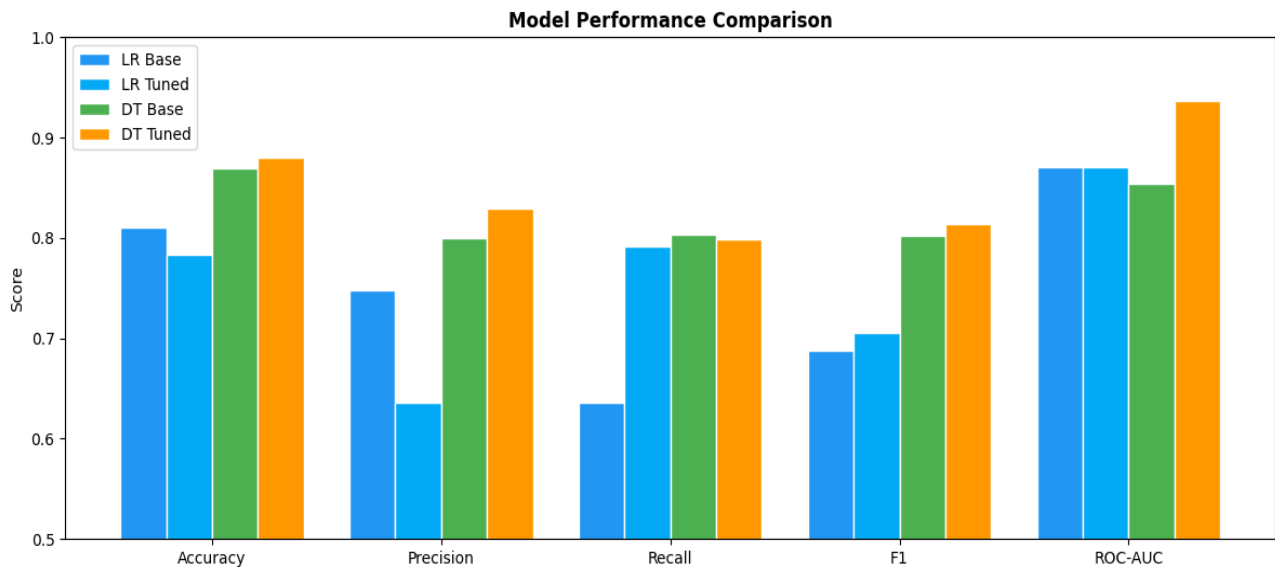


Figure 9: Side-by-Side Model Performance Comparison

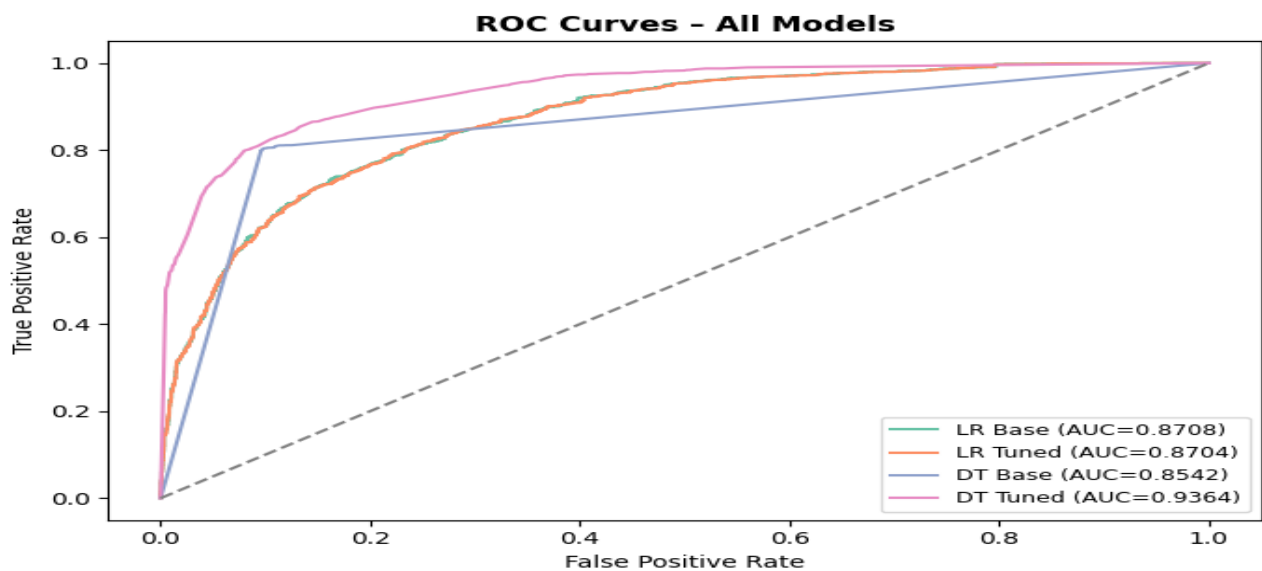


Figure 10: ROC Curves – All Four Models

Model Commentary:

- **LR Base:** Moderate performance. Good AUC (0.8708) but poor recall (63.61%) at default threshold – misses too many cancellations for business deployment.
- **LR Tuned:** Recall improves to 79.18% via optimal thresholding and backward elimination. AUC is comparable to base. Precision drops, but catching more cancellations is the priority.

- **DT Base:** High recall (80.40%) and accuracy but low AUC (0.8542) and extreme depth (33) indicate severe overfitting. Not reliable for generalisation.
- **DT Tuned:** Best overall model. AUC of **0.9364** (highest by far), accuracy 0.8804, recall 0.7989, F1 0.814. Depth reduced to 11 — balanced complexity and performance.

Final Model Selected: Tuned Decision Tree

The tuned decision tree (max_depth=11, min_samples_leaf=5) is selected as the final model because it achieves the highest ROC-AUC (0.9364), the best F1 score (0.814), strong recall (0.7989), and interpretable feature importances — directly actionable for hotel management.

7. Actionable Insights and Recommendations

1. Lead Time – Implement Tiered Cancellation Policies

Lead time is the single strongest predictor of cancellation. Bookings made more than 180 days in advance carry cancellation rates exceeding 50%. The hotel should implement non-refundable or partial-refund pricing tiers for bookings with long lead times. Offering a 5–10% discount for non-refundable commitments on bookings >90 days out could shift behaviour meaningfully.

2. Pricing – Dynamic Risk-Adjusted Pricing

Higher room prices are associated with higher cancellation probability. Revenue management should factor in cancellation risk when setting prices. Consider offering flexible rates at a premium and non-flexible rates at a slight discount — allowing guests to self-select and reducing revenue loss from last-minute room vacancies.

3. Special Requests – Increase Guest Engagement at Booking

Guests with zero special requests cancel far more frequently. The booking flow should actively prompt guests to add preferences (bed configuration, dietary needs, floor preference). Even a single special request increases psychological commitment and reduces cancellation intent significantly.

4. Online Segment – Stricter OTA Policies

Online bookings (primarily from OTAs) have the highest cancellation rate. INN Hotels should negotiate stricter cancellation windows with OTA partners, promote direct-booking incentives (best rate guarantee, free breakfast), and apply tighter deposit requirements for online channel bookings.

5. Loyalty Programme – Grow Repeat Guest Base

Repeated guests have near-zero cancellation risk. A structured loyalty programme offering points, priority check-in, and personalised offers would grow this segment and reduce portfolio-level cancellation risk over time.

6. Deploy the Predictive Model for Real-Time Scoring

The Tuned Decision Tree (ROC-AUC = 0.9364) should be deployed in the reservations system to score each booking at the time of confirmation. Bookings scoring above the model threshold should trigger a retention workflow: a personalised confirmation email highlighting the value of the stay, an early check-in offer, or a follow-up call from reservations. Estimated uplift: recovering 10–15% of high-risk bookings would translate directly to improved RevPAR.

Summary

This analysis built and compared four machine learning models to predict hotel booking cancellations for INN Hotels Group. The tuned Decision Tree classifier emerged as the best model across all key metrics. The most important features driving cancellation were lead_time, avg_price_per_room, online booking channel, special requests, and arrival month. The recommendations presented — tiered pricing, OTA policy tightening, loyalty investment, and real-time model deployment — provide a clear roadmap for the hotel group to meaningfully reduce revenue loss from cancellations.